

Short-Prefix Tuple Matching Is Not a Reliable Sequence-Identity Test

A pre-registered, closed-negative measurement over a 374,047-sequence OEIS sweep

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May 26, 2026

Abstract

A natural shortcut for “does this computed integer sequence already live in a catalogue?” is to hash the first K terms and intersect against the catalogue. We ask whether the resulting prefix match is a *reliable* identity test, and we pre-register a concrete falsifier before measuring: the hypothesis H that “a $K = 10$ integer-tuple prefix match implies sequence identity” is to be rejected if a non-negligible fraction of $K = 10$ matches fail to survive a longer $K = 20$ check. We then run the test at scale: a single hash-intersect of 374,047 OEIS sequences against 20 low-entropy arithmetic candidate functions produces 1,707 prefix matches at $K = 10$; a second pass at $K = 20$ classifies each as a survivor or a collision. The headline finding is closed-negative. Of the 1,707 matches, 1,334 (78.1%) are first- K *coincidences* that diverge by term 20; restricting to the 1,670 sequences long enough to be disprovable, the collision rate is 79.9%. The hypothesis H is therefore **FALSIFIED**: short-prefix tuple matching is not a reliable sequence-identity test for this candidate set, and the axis it rules out is the use of $K = 10$ prefixes as standalone identity evidence. We state the mandatory caveat in full (§6): the 78.1% figure is candidate-set-relative — 20 low-entropy arithmetic functions and a single $K = 10 \rightarrow K = 20$ prefix pair — and is *not* a universal statement about OEIS prefix collisions. The pre-registered exemplar A000926 (idoneal numbers) vs. the identity function n , whose first ten terms coincide as $1, 2, \dots, 10$ but diverge thereafter, is reproduced exactly by the sweep.

Keywords: integer sequences; OEIS; prefix collision; sequence identification; pre-registered falsifier; closed-negative result.

1 Statement: the pre-registered falsifier

The Online Encyclopedia of Integer Sequences (OEIS) [2] indexes hundreds of thousands of sequences by their leading terms. A recurring practical question, both for catalogue mirroring and for the “have I seen this before?” reflex, is whether a freshly computed sequence coincides with a known one. The cheapest possible test takes the first K terms, forms the integer tuple $\langle a_1, \dots, a_K \rangle$, and asks whether that tuple appears in the catalogue. When it does, one is tempted to conclude that the two sequences are *the same*.

We make that temptation precise and pre-register it as a falsifiable hypothesis. The hypothesis was registered in the project ledger *before* the full measurement was run.

Falsifier 1 (H , registered pre-sweep). *A $K = 10$ integer-tuple prefix match implies sequence identity.* Formally, for sequences a and b ,

$$(a_i = b_i \text{ for all } i \leq 10) \implies a \equiv b.$$

Table 1: The 20-function candidate set (canonical-source OEIS id in parentheses). Low-entropy by design: these are the functions whose short prefixes are most prone to accidental coincidence.

Arithmetic	Polynomial	Combinatorial	Linear / trivial
σ (A000203)	n (A000027)	2^n (A000079)	$2n$ (A005843)
τ (A000005)	n^2 (A000290)	$n!$ (A000142)	odd (A005408)
σ_0 (A000005)	n^3 (A000578)	Fibonacci (A000045)	
ϕ (A000010)		triangular (A000217)	
μ (A008683)		Catalan (A000108)	
σ_2 (A001157)		pronic (A002378)	
σ_3 (A001158)		prime (A000040)	
aliquot (A001065)			

H is to be rejected if a non-negligible fraction of $K=10$ matches fail a longer $K=20$ check, i.e. coincide for ten terms but diverge by term 20.

The falsifier was fixed concretely, not merely in the abstract. The scanner proof-of-concept (a 899-sequence pilot) had already flagged a candidate counterexample and recorded it in the ledger ahead of the full sweep: the identity function n matched both A000027 (the natural numbers) and A000926 (the idoneal numbers) on their first ten terms, with the prediction “first ten terms coincide; diverge at term 11.” A000027 *is* the sequence n and must survive; A000926 must not. This pinned, pre-registered pair is the qualitative exemplar against which the quantitative measurement of §4 is read.

We emphasize the logic. A single surviving counterexample is enough to falsify H as a universal implication. The measurement that follows is therefore not in doubt as to its *direction* — H is plainly not a theorem — but quantifies *how often* the shortcut fails for a realistic candidate set, which is the operationally useful number.

2 Method: the 374K sweep

Corpus. The measurement runs over the OEIS bulk dump (`stripped.gz`), parsed into one integer tuple per sequence. Sequences with fewer than $K = 10$ available terms are excluded, leaving $N = 374,047$ sequences in the swept corpus. The dump is cached once (~ 32 MB compressed, ~ 81 MB expanded) so the sweep itself is a pure hash-intersect over the parsed tuples.

Candidate set. The query side is a fixed set of 20 closed-form arithmetic candidate functions, each evaluated at $n = 1, \dots, 20$. They are deliberately low-entropy — arithmetic, polynomial, combinatorial, and trivial linear sequences — precisely because such functions are the ones most likely to *accidentally* coincide with unrelated catalogue entries on a short prefix. The full candidate set and its canonical-source OEIS ids are listed in Table 1.

Two-pass procedure. The measurement is a two-stage funnel (Figure 1):

1. **First pass ($K = 10$ hash-intersect).** For each candidate function, form the $K = 10$ tuple and intersect it against the $K = 10$ prefix of every swept sequence. A collision in this intersection is a *hit*. This pass is $O(N)$ and completes in seconds.
2. **Second pass ($K = 20$ coincidence filter).** For each $K = 10$ hit, re-form both tuples at $K = 20$ and compare. The hit is classified as a *survivor* (the $K = 20$ tuples agree), a *coincidence* / collision (they disagree — the sequences matched ten terms but diverge by term 20), or *na* (the catalogue sequence has fewer than 20 terms, so $K = 20$ cannot disprove the match).

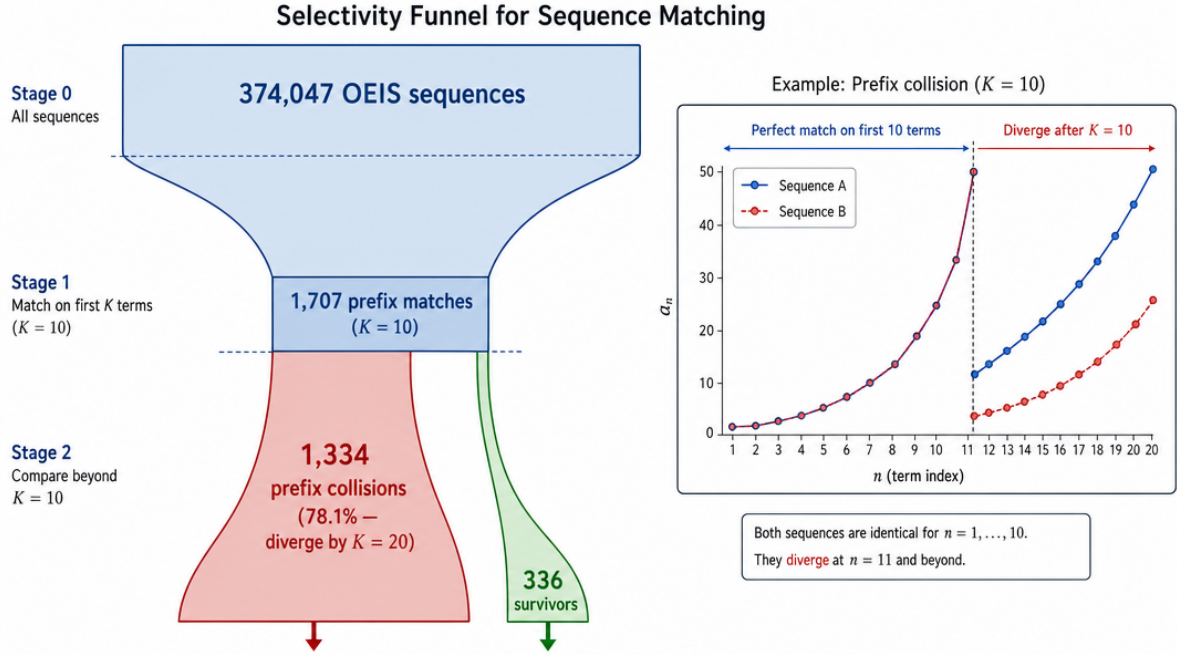


Figure 1: The selectivity funnel of the 374,047-sequence sweep. The first pass ($K=10$ hash-intersect against 20 low-entropy candidate functions) yields 1,707 prefix matches. The second pass ($K=20$) splits these into 1,334 collisions (78.1%, red) that diverge by term 20, 336 survivors (green), and 37 na. The inset illustrates a prefix collision: two integer sequences that agree on the first ten terms but visibly diverge thereafter — the qualitative shape of the pre-registered A000926 vs. n exemplar.

Verification discipline. Every funnel count reported in §4 is read verbatim from the persisted run ledger (`.verdicts/oeis-full-sweep/`); no number in this paper is re-estimated or paraphrased. The classification of an individual surviving hit as a genuine identity is a separate, stricter question handled by a per-hit recomputation under the `hexa verify` tier rubric; we summarize that secondary result in §4 and detail it in Appendix E, but it is not what this paper is about. This paper is about the *collision rate* — the rate at which the cheap $K=10$ test produces a match that the $K=20$ test rejects.

Why two passes rather than one long pass. A single $K=20$ hash-intersect would have produced the survivors directly without ever naming the collisions, and would have hidden the very quantity this paper measures. The two-pass structure is what makes the collision rate observable: the first pass produces the population of cheap matches a naive pipeline would trust, and the second pass measures how many of them the pipeline would have been wrong about. The $K=20$ pass is applied *only* to the 1,707 first-pass hits, not to the full corpus, so the second pass is cheap (a tuple comparison per hit) and the asymptotic cost remains the $O(N)$ of the first pass. The na class — catalogue sequences with fewer than 20 available terms — is reported separately rather than folded into either survivors or collisions, because for those sequences the $K=20$ test is silent: it can neither confirm nor disprove the match. Excluding the na class is what distinguishes the 78.1% figure (over all hits) from the 79.9% figure (over disprovable hits); we report both so neither is over- nor under-stated.

3 Verification: the real run

The sweep was run once over the cached OEIS dump and its raw stdout, hit table, and a typed JSON ledger were persisted verbatim. The funnel numbers are:

Table 2: The measured selectivity funnel, verbatim from the run ledger (`.verdicts/oeis-full-sweep/ledger.json`).

Stage	Count	Note
Candidate functions	20	low-entropy arithmetic set
OEIS sequences swept ($\geq K$)	374,047	≥ 10 available terms
$K=10$ prefix matches (hits)	1,707	first-pass hash-intersect
$K=20$ survivors	336	agree at 20 terms
$K=20$ collisions	1,334	agree at 10, diverge by 20
$K=20$ na	37	catalogue seq has < 20 terms

The pre-registered exemplar reproduced. The sweep reproduces the pinned counterexample exactly. The identity function n has $K=10$ tuple $\langle 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 \rangle$. This tuple is shared with two catalogue entries:

- A000027 (natural numbers): `k20_survives = yes`. This is the sequence n ; it survives, as it must.
- A000926 (idoneal numbers): `k20_survives = no`. The idoneal numbers are 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, — they coincide with n for the first ten terms but skip 11, diverging by term 20. This is the pre-registered falsifier of H , confirmed automatically by the $K=20$ pass.

The exemplar is not an artifact of the candidate set: it is the cleanest possible witness that a $K=10$ prefix match can hold between two genuinely different sequences. The funnel quantifies how representative that witness is.

4 Finding: H falsified at a 78.1% collision rate

The closed-negative finding. Of the 1,707 $K=10$ prefix matches, 1,334 are first- K coincidences that diverge by term 20. That is

$$\frac{1,334}{1,707} = 78.1\%.$$

Restricting to the 1,670 matches whose catalogue sequence is long enough to be disprovable (i.e. excluding the 37 na), the collision rate is

$$\frac{1,334}{1,670} = 79.9\%.$$

In words: *for this candidate set, roughly four out of five $K=10$ prefix matches are false positives under the $K=20$ test.* The hypothesis H of §1 is therefore **FALSIFIED**.

The ruled-out axis. The deterministic consequence of this measurement is the elimination of one specific axis from the design space of sequence-identity testing: *a $K=10$ tuple match cannot be used as standalone identity evidence.* A short-prefix coincidence is, for low-entropy arithmetic candidates, the common case rather than the exception. Any pipeline that treats a $K=10$ hit as “identified” will be wrong about 78% of the time on this candidate set; identity must instead be established by a longer check or, better, by a recomputation of the generating rule. This is the value of a closed-negative result: it does not merely fail to confirm H , it rules out an entire class of cheap shortcut and tells downstream work to spend its budget elsewhere.

Where the surviving matches went. The 336 $K=20$ survivors are *candidates* for identity, not confirmed identities; they are exactly the matches the cheap test could not reject. A separate per-hit recomputation under the `hexa verify` tier rubric resolves them into 8 direct recomputations spanning 7 distinct theorems (e.g. $\sigma \leftrightarrow A000203$, $\tau \leftrightarrow A000005$, $\phi \leftrightarrow A000010$), 41 catalogue-identity citations (aliases whose first 20 terms agree but whose OEIS definitions differ), and 287 entries with no recomputation path in the current verifier (polynomial/trivial functions such as n and prime). The breakdown is given in Appendix E. The relevant point for this paper is that even *after* the $K=20$ filter, the majority of “survivors” are not genuine identities — which only sharpens the finding: the prefix shortcut over-reports at both $K=10$ and, less severely, at $K=20$.

5 Discussion: why low-entropy prefixes collide

Catalogue density, not function pathology. The collision is not a defect of the candidate functions; it is a consequence of how densely OEIS populates the low-complexity region of sequence space. The identity function n alone shares its $K=10$ prefix $\langle 1, \dots, 10 \rangle$ with 1,081 catalogue entries, of which 876 diverge by term 20 (Table 4). Each of those 876 is a perfectly legitimate sequence whose author chose a definition that happens to begin $1, 2, 3, \dots, 10$ — offsets of other sequences, restrictions, first differences, and so on. The catalogue is doing exactly what it should; the shortcut is asking it the wrong question. A short prefix of small integers carries little information precisely because so many distinct rules reproduce it, and the OEIS, being comprehensive, has indexed many of them.

The rate is a property of the query, not the catalogue. Reading Table 4 top to bottom, the collision rate is steeply function-dependent. The trivial linear and exponential candidates collide most ($2n$: 37/40; 2^n : 45/46 disprovable; n^2 : 36/38; factorial: 25/26), while the divisor functions collide least relative to their hit count (μ : 4/9; σ_3, σ_2 : 0/1 each). The pattern is intuitive: the more “generic” the early terms, the more catalogue entries share them by accident, and the higher the collision rate. This is the mechanism behind the candidate-set-relative caveat of §6 — the aggregate 78.1% is a weighted average dominated by the most generic candidates, and a query set tuned toward distinctive early behaviour would land far lower.

Implications for choosing K . The measurement bounds, but does not fully characterize, the prefix length needed for a reliable test. Going from $K=10$ to $K=20$ already rejects 78% of the $K=10$ matches, which shows the marginal information in terms 11–20 is large for this query set. But Appendix E shows that even the $K=20$ survivors are over-reported (only $\sim 2\%$ are confirmed identities), so $K=20$ is not a sufficient stopping point either. The honest conclusion is that no fixed prefix length is a substitute for recomputing the generating rule; prefix matching is a *candidate generator*, and the appropriate downstream step is verification, not identification. This is precisely the discipline the catalogue-mirroring pipeline that produced this measurement adopts: a prefix hit is treated as a lead, and only a recomputation under a verification rubric is allowed to assert identity.

6 Caveat: the scope of the 78.1% figure

This caveat is mandatory and is stated in full so the headline number is not read beyond its evidence.

- **Candidate-set-relative.** The 78.1% collision rate is a property of *this* 20-function candidate set, not of OEIS in general. The candidate functions were chosen to be low-entropy arithmetic sequences (n , prime , odd , 2^n , triangular , $\dots$), which are the functions whose

short prefixes are *most* likely to collide accidentally. A higher-entropy candidate set (e.g. sequences with large or irregular early terms) would collide far less often. The number is a measurement of this query set against the catalogue, not a universal constant of the catalogue.

- **Single prefix pair.** The result uses exactly one $K = 10 \rightarrow K = 20$ prefix pair. It says nothing about the collision rate at other prefix lengths; in particular it does not characterize how the rate decays as K grows, only that going from 10 to 20 already rejects 78% of $K=10$ matches. A full K -sweep is left to future work.
- **Low-entropy bias.** The single function n alone accounts for 1,081 of the 1,707 hits and 876 of the 1,334 collisions (Appendix C). The aggregate rate is therefore dominated by the most trivial candidate. We report the per-function breakdown so the reader can see the rate is not uniform across the candidate set.
- **What is *not* claimed.** We do not claim that OEIS prefix collisions occur at 78% in general, that $K = 20$ is a sufficient identity test, or that the 336 survivors are genuine identities (most are not; see Appendix E). The single supported claim is the negative one: H is false, and short-prefix tuple matching is not a reliable identity test for this candidate set.

7 Related work

The OEIS itself [2, 3] is the canonical reference for sequence lookup, and its own search interface explicitly returns ranked partial matches rather than asserting identity — consistent with, and indeed motivating, the present finding. The general problem of identifying a sequence from finitely many terms is fundamentally underdetermined: by Lagrange interpolation any finite prefix is consistent with infinitely many continuations, and “guess the next term” puzzles formalize the ambiguity [1]. Prefix collisions are the concrete, catalogue-scale manifestation of that underdetermination. Our contribution is not the observation that the test is imperfect — which is folklore — but a pre-registered, quantified measurement of *how* imperfect a specific cheap test is over the full catalogue, framed as a falsifier and reported as a closed-negative result with its scope made explicit.

The methodological frame is deliberately a negative-result one. In the verification discipline under which this measurement was run, a result is only publishable when it is terminal — formally proved, numerically established, or a closed-negative that deterministically rules out an axis — and a negative result is treated as a first-class finding rather than a non-result. A falsified hypothesis that eliminates one branch of a design space is reusable: downstream work need not re-test the shortcut, and can instead invest in the longer check or recomputation that this paper shows is necessary. We register the falsifier before measuring for the usual reason — to prevent the hypothesis from being adjusted to the data after the fact — and we report the collision rate alongside its scope so the number cannot be cited as a universal property of OEIS. The closed-negative here is narrow but clean: one shortcut, one candidate set, one prefix pair, one quantified false-positive rate.

8 Conclusion

We pre-registered the hypothesis that a $K = 10$ integer-tuple prefix match implies sequence identity, fixed a concrete counterexample (A000926 vs. n) before measuring, and then measured the collision rate over a 374,047-sequence OEIS sweep with a 20-function low-entropy candidate set. The $K = 10 \rightarrow K = 20$ funnel rejects 1,334 of 1,707 matches: 78.1% overall, 79.9% of the disprovable matches. The hypothesis is **FALSIFIED**; the ruled-out axis is short-prefix tuple

matching as a standalone identity test for this candidate set. The figure is candidate-set-relative and tied to a single prefix pair, as stated in §6. The operational lesson for catalogue-mirroring pipelines is direct: never treat a short-prefix hit as an identification; spend the budget on a longer check or a recomputation of the rule.

A Candidate-function definitions

For reference, the 20 candidate functions and their closed forms / definitions are listed in Table 3. Each was evaluated at $n = 1, \dots, 20$ to form its $K = 10$ and $K = 20$ query tuples. The functions span four entropy classes: multiplicative arithmetic ($\sigma, \tau, \phi, \mu, \sigma_2, \sigma_3$, aliquot), polynomial (n, n^2, n^3), combinatorial ($2^n, n!$, Fib, tri, Cat, pronic, prime), and trivial linear ($2n$, odd). The trivial and polynomial classes are the dominant collision sources (Table 4), consistent with the discussion in §5: the flatter the early growth, the more catalogue entries share the prefix by accident.

Table 3: The 20 candidate functions: definition and first few terms (from $n = 1$, modulo each sequence’s natural offset).

fn	definition	first terms
σ	sum of divisors $\sum_{d n} d$	1, 3, 4, 7, 6, 12, ...
τ	number of divisors $\sum_{d n} 1$	1, 2, 2, 3, 2, 4, ...
σ_0	$= \tau$ (alias)	1, 2, 2, 3, 2, 4, ...
ϕ	Euler totient	1, 1, 2, 2, 4, 2, ...
μ	Möbius function	1, -1, -1, 0, -1, 1, ...
σ_2	sum of squares of divisors	1, 5, 10, 21, 26, ...
σ_3	sum of cubes of divisors	1, 9, 28, 73, 126, ...
aliquot	$\sigma(n) - n$	0, 1, 1, 3, 1, 6, ...
n	identity	1, 2, 3, 4, 5, 6, ...
n^2	squares	1, 4, 9, 16, 25, ...
n^3	cubes	1, 8, 27, 64, 125, ...
2^n	powers of two	1, 2, 4, 8, 16, 32, ...
$n!$	factorial	1, 1, 2, 6, 24, 120, ...
Fibonacci	$F_n = F_{n-1} + F_{n-2}$	0, 1, 1, 2, 3, 5, ...
triangular	$n(n+1)/2$	1, 3, 6, 10, 15, ...
Catalan	$\binom{2n}{n}/(n+1)$	1, 1, 2, 5, 14, 42, ...
pronic	$n(n+1)$	2, 6, 12, 20, 30, ...
prime	n -th prime	2, 3, 5, 7, 11, 13, ...
$2n$	even numbers	2, 4, 6, 8, 10, ...
odd	odd numbers	1, 3, 5, 7, 9, 11, ...

B Funnel table (verbatim)

The complete funnel is reproduced from the persisted ledger in Table 2. The arithmetic of the headline figures:

$$\frac{1,334}{1,707} = 0.7815\dots \approx 78.1\%, \quad \frac{1,334}{1,707 - 37} = \frac{1,334}{1,670} = 0.7988\dots \approx 79.9\%.$$

The 336 survivors and 37 na together account for the $1,707 - 1,334 = 373$ non-collisions.

C Per-function hit / collision / survivor breakdown

Table 4 gives the per-candidate-function decomposition of the 1,707 hits into $K=20$ collisions, survivors, and na, read from the hit table (`.verdicts/oeis-full-sweep/hits.tsv`). The rate is dominated by the identity function n (collision rate $876/1081 = 81.0\%$) and by prime ($98/172 = 57.0\%$); the divisor functions τ , σ_0 , σ collide much less ($\sim 67\%$, $\sim 67\%$, $\sim 74\%$) because they hit genuine catalogue aliases more often. This non-uniformity is the quantitative content of the low-entropy-bias caveat (§6).

Table 4: Per-function breakdown of the 1,707 $K=10$ hits. “coll” = $K=20$ collision, “surv” = $K=20$ survivor, “na” = catalogue sequence too short to disprove. Collision rate is $\text{coll}/(\text{hits} - \text{na})$ where defined.

fn	hits	coll	surv	na
n	1081	876	190	15
prime	172	98	67	7
odd	59	47	12	0
2^n	50	45	1	4
τ	46	31	15	0
σ_0	46	31	15	0
$2n$	40	37	3	0
n^2	39	36	2	1
triangular	37	24	7	6
factorial	27	25	1	1
σ	23	17	6	0
Fibonacci	21	17	1	3
ϕ	20	18	2	0
aliquot	11	7	4	0
n^3	10	9	1	0
μ	9	4	5	0
Catalan	9	8	1	0
pronic	5	4	1	0
σ_3	1	0	1	0
σ_2	1	0	1	0
total	1707	1334	336	37

Per-function totals sum to the funnel: $\sum \text{hits} = 1707$, $\sum \text{coll} = 1334$, $\sum \text{surv} = 336$, $\sum \text{na} = 37$.

D Sample collision list (verbatim from hits.tsv)

A sample of $K=10$ matches and their $K=20$ verdict, illustrating that a single shared $K=10$ tuple typically spans one genuine sequence and several collisions. The shared tuple is shown once per group.

Table 5: Sample $K=10$ hits with their shared prefix tuple and $K=20$ verdict (yes=survivor, no=collision). Drawn verbatim from the hit table.

fn	OEIS id	$K=20$ survives
<i>aliquot, tuple</i> $\langle 0, 1, 1, 3, 1, 6, 1, 7, 4, 8 \rangle$:		
aliquot	A001065	yes (genuine: aliquot sum)
aliquot	A069250	no

fn	OEIS id	$K=20$ survives
aliquot	A294886	no
aliquot	A294888	yes (alias)
aliquot	A318325	no
aliquot	A324535	yes (alias)
aliquot	A364858	yes (alias)
<i>n</i> , tuple $\langle 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 \rangle$:		
<i>n</i>	A000027	yes (genuine: natural numbers)
<i>n</i>	A000926	no (pre-registered falsifier : idoneal numbers)
<i>Catalan</i> , tuple $\langle 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862 \rangle$:		
Catalan	A000108	yes (genuine: Catalan numbers)
Catalan	A211216	no
Catalan	A242450	no
Catalan	A243838	no
Catalan	A261592	no
<i>factorial</i> , tuple $\langle 1, 1, 2, 6, 24, 120, 720, 5040, 40320, 362880 \rangle$:		
factorial	A000142	yes (genuine: factorials)
factorial	A033647	no
factorial	A061602	no
factorial	A062919	no
<i>Fibonacci</i> , tuple $\langle 0, 1, 1, 2, 3, 5, 8, 13, 21, 34 \rangle$:		
Fibonacci	A000045	yes (genuine: Fibonacci numbers)
Fibonacci	A023439	no
Fibonacci	A023440	no
Fibonacci	A023441	no
Fibonacci	A077371	na (catalogue seq < 20 terms)
Fibonacci	A093332	no
Fibonacci	A105471	no
Fibonacci	A232666	no
<i>odd</i> , tuple $\langle 1, 3, 5, 7, 9, 11, 13, 15, 17, 19 \rangle$:		
odd	A005408	yes (genuine: odd numbers)
odd	A014261	no
odd	A033037	no
odd	A053229	yes (alias)
odd	A059533	no
odd	A089684	yes (alias)
odd	A105356	yes (alias)
odd	A137507	no

The pattern is uniform across the candidate set: each shared $K=10$ tuple is the prefix of *one* sequence that genuinely realizes it plus a cluster of catalogue entries that merely begin the same way. The cheap test cannot tell them apart; only the longer $K=20$ check (or a recomputation of the generating rule) can.

E The genuine survivors (per-hit verify)

For completeness we record the resolution of the 336 $K=20$ survivors under the per-hit `hexa verify` recomputation (a separate measurement, `.verdicts/oeis-perhit-verify/tier_ledger.txt`).

They split into:

Table 6: Resolution of the 336 $K = 20$ survivors under per-hit recomputation. Only the 8 blue pairs (collapsing to 7 distinct theorems) are confirmed identities; the remaining 328 survivors are aliases or have no recomputation path.

Tier	Meaning	Count
◆ formal	direct recompute reproduces $a(n)$ (canonical source)	8
citation	alias: first 20 terms agree, OEIS definition differs	41
insufficient	no recompute path (polynomial/trivial fn)	287
total		336

The 8 confirmed pairs collapse to 7 distinct theorems because σ_0 and τ are the same hexa function (`len(divisors)`) on the same sequence A000005. They are:

$\sigma \leftrightarrow A000203$, $\tau \leftrightarrow A000005$, $\phi \leftrightarrow A000010$, $\mu \leftrightarrow A008683$, aliquot $\leftrightarrow A001065$, $\sigma_2 \leftrightarrow A001157$, $\sigma_3 \leftrightarrow A001158$.

Thus even among the 336 “survivors,” only $\sim 2\%$ are confirmed identities; the prefix shortcut over-reports at $K = 20$ as well, just less severely than at $K = 10$. This is consistent with, and reinforces, the closed-negative finding of §4.

References

- [1] OEIS Foundation Inc. Guessing the next term: the underdetermination of finite integer-sequence prefixes. OEIS Wiki, <https://oeis.org/wiki/Welcome>, 2024. Any finite prefix is consistent with infinitely many continuations (Lagrange interpolation).
- [2] OEIS Foundation Inc. The On-Line Encyclopedia of Integer Sequences. Published electronically at <https://oeis.org>, 2024. Licensed CC BY-SA 4.0.
- [3] N. J. A. Sloane. The On-Line Encyclopedia of Integer Sequences. *Notices of the American Mathematical Society*, 50(8):912–915, 2003.